AI in Supply Chain & Logistics

Project Report

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1. Executive Summary

A large Indian logistics company is losing money and goodwill for three connected reasons. Parcels reach customers late, fuel takes a big share of running costs, and demand swings so sharply that planners spend their time reacting instead of planning. This report explains how artificial intelligence (AI) can address each problem and how the three fixes can work together.

Our recommendation has three parts. First, replace fixed routes with a routing system that plans and re-plans using live traffic and weather. Second, build a demand forecast that predicts parcel numbers for each city and day, using festivals, sale dates and past orders. Third, bring AI into the warehouses to decide where goods are kept, how workers and robots pick them, and when trucks are loaded. We suggest a phased rollout over 18 months, starting with routing because fuel savings show up fastest and can help pay for the later phases. Supporting tables and charts are in the Appendix. All figures in this report are planning targets, not results from a real pilot.

2. Background

Indian logistics moves goods over long distances, across very different roads and weather, to customers who now expect fast and trackable delivery. Online shopping has raised these expectations sharply, and festival periods such as Diwali can bring volumes several times higher than a normal week. A company working here has to deal with traffic in big cities, toll plazas and state borders on highways, heavy rain in the monsoon, and addresses that are often written with landmarks instead of exact street numbers.

For the company in this case, the effect shows up in three places. Late deliveries lead to complaints, penalties from business clients and lost repeat orders. Fuel is one of the largest running costs for any fleet, so a small waste per vehicle becomes a very large amount across thousands of vehicles. And when demand is hard to predict, the company either keeps too many vehicles and staff waiting or hires extra trucks in a hurry at high prices. The National Logistics Policy of 2022 aims to make the whole sector cheaper and more efficient, so companies that improve early are likely to gain an advantage.

3. Why These Problems Happen

Before choosing tools, it helps to be clear about the causes. Delays usually come from fixed routes that ignore real conditions, slow loading and sorting at hubs, and failed first attempts caused by wrong addresses or customers who are not available. High fuel cost comes from routes that are longer than needed, trucks returning empty, engines left running at docks and tolls, and driving habits such as sudden braking. Unpredictable demand comes from forecasts built on last year’s numbers and the planner’s experience, which cannot capture the combined effect of sales events, weather and local trends.

These causes are linked. A poor forecast leads to poor planning at the hub, which delays dispatch, which forces drivers to rush. Table A1 in the Appendix connects each problem to its cause and to the AI response we propose.

4. Using AI to Optimise Routing

The first step is to replace fixed daily routes with a system that chooses routes based on what is actually happening. The AI takes in the day’s orders, the size and load of each vehicle, delivery time slots, driver working hours, and live information on traffic, road closures and weather. It then works out which vehicle should carry which parcels and in what order to make the stops, so that total distance and time are as low as possible. A human planner can still review the plan, but the system does the heavy calculation that nobody could do by hand for thousands of vehicles.

The bigger gain comes after the vehicle leaves the hub. If an accident blocks a highway or a sudden downpour slows a city, the system recalculates and sends a new route to the driver’s phone within minutes. It can also suggest moving a few stops from a late vehicle to one nearby. Over time the system learns how long trips really take on each road at each hour, which makes the estimated arrival time far more honest than a standard rule of thumb.

Fuel savings come from several places at once. Shorter routes are the obvious one. Matching return trips with nearby loads is another, because a truck that travels back empty burns fuel without earning anything. GPS trackers and engine sensors also show idling time, harsh braking and speeding. The AI turns this into a simple score for each driver along with practical tips, and when this is paired with fair incentives it often helps change habits. The same data can warn when a vehicle is likely to need repair, so it can be fixed in the depot instead of breaking down on the road, which costs both fuel and time. Fig. A1 in the Appendix shows the fuel cost path we are aiming for.

India needs one extra step that many global systems skip, which is address cleaning. Addresses often mix landmarks, local names and spelling differences, and a wrong pin point sends the driver to the wrong lane. AI can read these addresses, correct spellings, and fix the delivery point using the history of earlier successful deliveries. This reduces failed first attempts, one of the biggest hidden causes of wasted kilometres. To work well, the routing system needs GPS data from every vehicle, order data, FASTag toll records, and map and traffic data (see Table A2).

5. Using AI to Predict Demand

Demand prediction means estimating how many parcels will need to move on which route on which day. Today this is often done with last year's numbers plus judgement. An AI model can look at many factors together: past orders by pin code, the festival and sale calendar, wedding season, weather, customer promotions and local events. It learns which combinations lead to spikes and gives a forecast with a range, for example "between 9,000 and 11,000 parcels for this city tomorrow".

The forecast should be detailed: by city or hub, by route and by day, and not one number for the whole country. A short-term forecast is refreshed every day, and a longer view of four to eight weeks is used to plan vehicles and staff. Once this is in place, many decisions change. Parcels can be moved closer to areas where orders are about to rise. Contract vehicles can be booked early at agreed rates instead of being hired at peak prices. Warehouse shifts can be planned to match the work, which reduces both overtime and idle time. Fig. A3 shows the improvement in forecast accuracy we expect compared with current methods.

Planners stay in control. If a planner knows that a large customer is about to run a promotion that the data does not yet show, they can adjust the forecast, and the model learns from the correction. Every week the team should compare predicted and actual volumes and review the biggest misses. This habit matters more than the choice of model, because it keeps the forecast honest and builds trust among the people who use it.

6. Using AI to Automate Warehouse Operations

Warehouses are where time is easily lost, and a late start at the hub means a late delivery on the road. AI can help in four practical ways. The first is smart placement: fast-moving items are kept near the dispatch gates and slow movers further away, and the AI updates this as demand changes. The second is better picking: workers receive the shortest walking path through the aisles, so each order is picked faster and with less tiredness. The third is robots and sorting machines: small robots carry parcels, and automatic sorters send each parcel to the correct truck bay. The fourth is cameras with AI that read labels, count parcels and spot damage at the time of loading, which reduces mix-ups and wrong dispatches.

Two smaller changes also matter. AI can schedule truck arrival slots so that vehicles do not queue outside the warehouse with engines running, which saves time and fuel. And the demand forecast can tell managers how many people are needed in each shift.

Automation should be introduced in steps. In India, labour is available and full automation is expensive, so robots are best used where work is heavy, repetitive and time-critical, such as sorting at the busiest hubs. Workers can move into roles such as supervising machines, handling exceptions and quality checks. The company should pilot in two large warehouses first, measure the gain in parcels handled per hour, and expand only after that. This approach also reduces fear among staff, which is a real risk and is covered in Section 11.

7. Improving Customer Satisfaction

Efficiency matters to the company, but customers judge only what they experience. They want the parcel on time, they want to know where it is, and they want problems solved quickly. AI supports each of these. Better routing and forecasting make delivery times more reliable, so the promise made at booking is kept more often. Live tracking with an honest arrival window replaces vague messages. Automatic SMS or WhatsApp alerts warn the customer when a delay is likely, which is far better than silence. A chat assistant that answers in Hindi, Gujarati and other regional languages can handle routine questions at any hour and pass difficult cases to a person. Finally, the system can predict which deliveries are likely to fail, for example when a customer has often been unavailable at that hour, and ask them to confirm or change the time slot before the vehicle leaves. Fewer failed attempts means lower cost and fewer unhappy customers at the same time.

8. One Connected System

The three solutions work best when they share data. The demand forecast tells the warehouse how much to prepare and tells the routing system how many vehicles will be needed. The warehouse system tells routing when parcels will actually be ready. The routing system tells customers when to expect delivery. A single control-tower screen that shows vehicles, hubs and delays in one place lets managers see problems early and act. Fig. A5 shows how the pieces connect. The systems could be built separately and joined later, but it is cheaper to agree on shared data from the start.

9. Implementation Plan

We propose three phases over about 18 months (Fig. A4). Phase 1, months 0 to 3, is the foundation: fit GPS trackers on every vehicle, clean and join the data from orders, trips and warehouses, and run a routing pilot on three or four busy routes. Phase 2, months 3 to 9, is scale-up: roll out smart routing on the main routes, launch the daily demand forecast, and start AI-based placement and picking routes in two large warehouses. Phase 3, months 9 to 18, is automation: add sorting machines and robots at the busiest hubs, build the control-tower screen, and extend the system to small-town and rural routes. At the end of each phase, results are checked against targets, and money for the next phase is released only if the results justify it.

10. Expected Benefits and How We Will Measure Them

We expect gains in cost, speed and customer experience. Table A4 lists the measures we will track, with indicative targets such as on-time delivery of 95 percent or more, fuel cost per km down by 8 to 12 percent, and a 20 to 30 percent fall in empty return trips. Table A3 gives a simple worked example: for a fleet of 2,000 vehicles, a 10 percent fuel saving is worth about ₹34 crore a year under the stated assumptions. Fig. A2 shows how on-time delivery could improve over the 18 months. These are planning targets and not guaranteed results, so the Phase 1 pilot must confirm what is realistic for this company.

11. Risks and How to Manage Them

The main risks can be managed if they are planned for (Table A5). Poor data is the most common cause of failure, so data cleaning comes first. Staff may worry about losing their jobs, so training and honest communication are needed from the start. Costs should be controlled by spending in phases. Mobile networks are weak on some rural routes, so driver apps must work offline. Customer data must be protected under India's Digital Personal Data Protection Act, 2023. Finally, the company should not trust the system blindly: a human should always be able to override it, and results should be reviewed every month.

12. Conclusion and Recommendation

AI can make this company's deliveries faster, cheaper and more reliable, but only if it is introduced in a planned way. Smart routing reduces fuel use and delay, accurate forecasting removes the guesswork that leaves trucks empty or hubs overloaded, and warehouse automation speeds up dispatch. Together they improve both efficiency and customer satisfaction. We recommend that management approve Phase 1 now, begin with routing on the busiest routes, and decide on the later phases once the first results are in.

Appendix: Tables and Charts

Table A1: Problems, causes and AI responses

Problem	Why it happens today	How AI helps	Expected result
Delayed deliveries	Fixed routes, traffic jams, bad weather, slow hubs, unclear addresses	Live route changes, honest arrival times, faster hub handling, address cleaning	More on-time deliveries, fewer complaints
High fuel cost	Long routes, empty return trips, engine idling, rough driving	Lowest-cost routes, matching return loads, driver coaching, early repair alerts	Lower fuel cost per km
Unpredictable demand	Forecasts based on last year's numbers and judgement; festival and sale spikes are missed	Forecast using past orders, festivals, weather and sale dates	Fewer last-minute truck hires and idle vehicles

Table A2: What the AI does and the data it needs

Use case	What the AI does	Data needed
Route planning	Chooses the best routes and order of stops	GPS, orders, traffic, weather, vehicle load
Live re-planning	Changes the route during the trip	Live traffic, road closures, driver location
Arrival-time prediction	Predicts when each parcel will arrive	Past trip times by road and hour
Address cleaning	Corrects addresses and fixes the drop point	Address history, delivery success records
Demand forecast	Predicts parcels by city, route and day	Past orders, festival and sale dates, weather
Warehouse placement and picking	Decides item location and picking path	Item movement history, warehouse layout
Camera checks	Reads labels, counts parcels, spots damage	Camera images of parcels
Dock scheduling	Gives each truck an arrival slot	Truck arrival and loading times

Table A3: Worked example of fuel saving (assumed numbers, for illustration only)

Item	Value
Fleet size	2,000 vehicles
Average distance per vehicle per day	250 km
Average mileage	4 km per litre
Diesel price	₹90 per litre
Working days per year	300
Fuel used per year	37.5 million litres
Fuel cost per year	₹337.5 crore
Saving at 10 percent	About ₹34 crore per year

Table A4: Measures of success and indicative targets

Measure	What it shows	Indicative target
On-time delivery	Share of parcels delivered on the promised day	95% or higher
Fuel cost per km	How efficiently vehicles are used	Reduce by 8–12%
Empty return trips	Share of kilometres run without a load	Reduce by 20–30%
Forecast error	Gap between predicted and actual parcels	Reduce by 20–30%
Parcels per warehouse hour	Speed of sorting and picking	Increase by 20–30%
Customer complaints	Late, lost or damaged parcel complaints	Reduce by 25% or more

Table A5: Main risks and responses

Risk	Chance	Effect	Response
Poor or missing data	High	High	Clean data first; begin with a small pilot
Staff resistance	Medium	High	Training, honest communication, new supervision roles
High cost	Medium	Medium	Spend in phases; use routing savings to fund the next phase
Weak network on rural routes	Medium	Medium	Driver apps that work offline and sync later
Customer data privacy	Low	High	Follow the DPDP Act, 2023; limit who can see data
Too much trust in AI	Medium	Medium	Human override at all times; monthly review of results

Charts (all figures are planning targets or assumed values, not measured results)

Fig. A1: Fuel cost per km (target path)

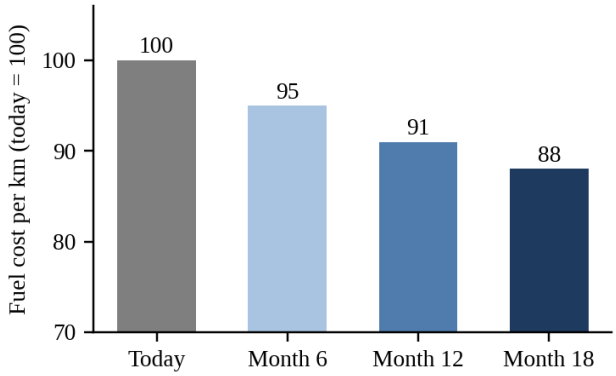


Fig. A2: On-time delivery (assumed start 88%)

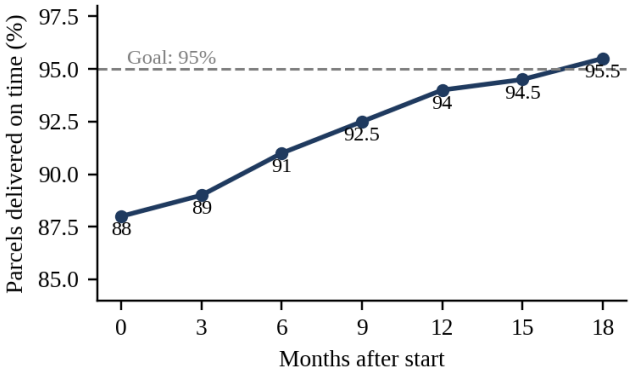


Fig. A3: Forecast accuracy (lower is better)

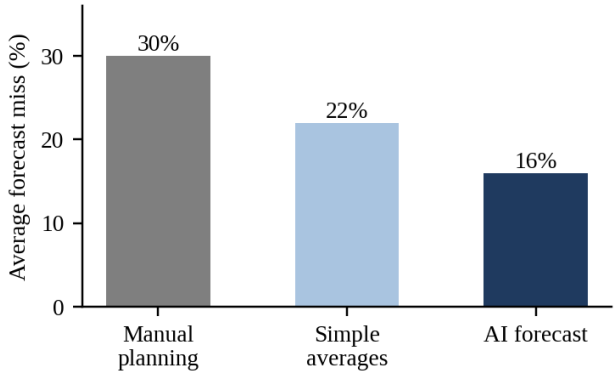


Fig. A4: 18-month roadmap

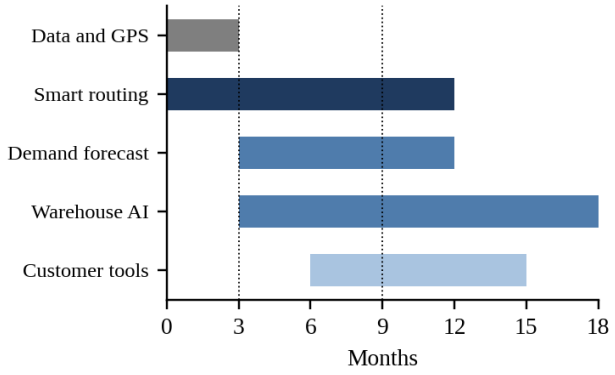


Fig. A5: How the three AI systems work together

